

Database

Without database:

- Storing data in text files, spreadsheets, etc.
- Which is a very slow process and retrieving data from a large, unstructured collections is error-prone, and inefficient.

Eg: Searching for a specific customer's order in 10,000 Excel files (takes hours)

With database

- Organize the collection of data in the form of table, graph, key-value pair, etc,...
- The task of storing, managing and retrieving the data becomes a easy task using tools like MySQL

Eg: Find the same customer's order in DB take less than a min with simple query

There are two different type of database

→ Relational database (structured i.e table)

→ Non-Relational database (unstructured i.e graph, doc)

Relational database: Stores data in the form of row and column, and have relationship among various table

Eg: MySQL, SQL server, PostgreSQL, etc..

Interview Question:

There are different types of databases right? There are Oracle, MySQL, ~~MySQL~~ MongoDB, SQL server, PostgreSQL.

What is the difference b/w ^{these} databases?

Non-Relational database:

Store ~~in the form of~~ data in the form of graph, key-value pair, doc.

Eg: HBase, MongoDB, Cassandra, etc

There are two different type of database

→ Relational database (structured i.e table)

→ Non-Relational database (unstructured i.e graph, doc)

Relational database: Stores data in the form of row and column, and have relationship among various table

Eg: MySQL, SQL Server, PostgreSQL, etc..

Interview Question:

There are different types of databases right? There are Oracle, MySQL, ~~MySQL~~ MongoDB, SQL server, PostgreSQL.

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Non-Relational database:

Store ~~in the form of~~ data in the form of graph, key-value pair, doc.

Eg: HBase, MongoDB, Cassandra, etc

I/Q Ans:

→ There are different types of database like Oracle, MySQL, MongoDB, etc. The main difference is that relational database, non-relational database.

→ where MySQL, Oracle comes under RDB then MongoDB comes under NRDB.

Data → refers collection of raw facts or fig
Eg: XYZ, orange

Information → Processed data is called information → orange colour, orange fruit

↳ it gives meaning to a unprocessed data

SQL

MySQL^{data} base → to handle the MySQL database, we need Query language called SQL.

→ MySQL is a tool on which we are writing a SQL query to fetch the data, store the data and also to manage the data.

✓ Practice: GooxMIDE platform (cloud platform)
based

✓ MySQL workbench

COMMANDS

CD1

SHOW DATABASES; (show all the available databases in the DB^s software)

CD2

CREATE DATABASE {techForAllWithMonish};

(this command is used to create a new database)

CD3 CREATE DATABASE IF NOT EXISTS {techForAllWithMonish}

(it only create database if it is not available in MySQL platform)

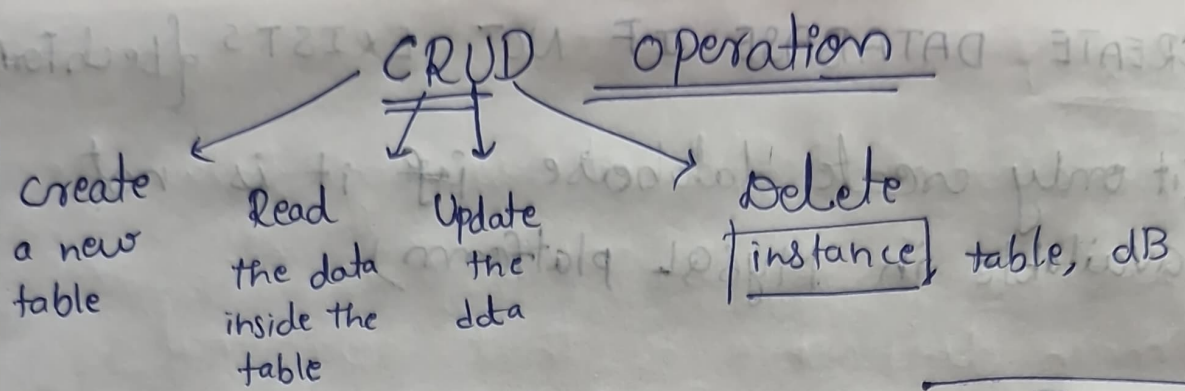
→ This query is used because, if we try to create the already existing DB with the command CREATE DATABASE {techForAllWithMonish}

Error → It will give error as DB already exists
→ if we use IF NOT EXISTS command: it will give warning

CD4 USE techForAllWithMonish;
(to choose particular database)

CD5 SELECT DATABASE();
(to identify the current DB that we are working)

[After choosing the ~~DB~~ only we can use CD5]



Creating a new table

INT
VARCHAR

SYNTAX:

CREATE TABLE {table-name} (

col 1 datatype,

col 2 datatype,

);

Eg

CREATE TABLE employee (

FirstName varchar(30),

LastName varchar(30),

Location varchar(20),

Salary int

);

to get Description of table

SYNTAX:

DESC {table_name}

Eg:

DESC employee;

o/p

<u>Field</u>	<u>Type</u>	<u>Null</u>	<u>key</u>	<u>Default</u>	<u>Extra</u>
FN	varchar(30)	YES		NULL	
LN	(40)	YES		NULL	
L	(20)	YES		NULL	
S	int	YES		NULL	